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Roll No.:

Amrita Vishwa Vidyapeetham

Missed Mid Term Examination, May 2024

Second Semester

Integrated MSc Physics

22MAT123 Differential Equations and Linear Algebra

Time: 2hrs

Maximum Marks: 50 marks

Answer all questions

$y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$

Qn. No.		Marks	CO	Blooms Taxonomy level
1	An experimental population of fruit flies increases according to the law of exponential growth. There were 100 flies after the second day of the experiment and 300 flies after the fourth day. Approximately how many flies were in the original population?	5	CO1	L2
2	Solve the first order differential equation $\frac{dy}{dx} = \frac{3}{xy^2}$, $y(1) = 3$.	5	CO1	L1
3	Check the exactness of the differential equation $(3y^2 + 4x)dx + 3xydy = 0$. If not exact, find an integrating factor and hence find the general solution.	5	CO1	L2
4	Verify that the functions $x, xlnx$ form a basis of solutions for the differential equation $x^2y'' - xy' + y = 0$. If so, solve the equation with respect the initial value conditions $y(1) = 4.3$, $y'(1) = 0.5$.	5	CO2	L2
5	Solve the initial value problem $8y'' - 2y' - y = 0$, $y(0) = -0.2, y'(0) = -0.325$.	5	CO2	L3
6	Find a general solution $y(x)$ to the Euler-Cauchy equation $x^2y'' - 3xy' + 4y = 0$. If $y(1) = -2$, $y'(1) = -7$, then find a particular solution.	5	CO2	L3

7	Let the functions $y_1(x) = e^{-x}cos3\pi x$ and $y_2(x) = e^{-x}sin3\pi x$ forms solutions to a linear differential equation. Then (a) check their linear independence by Wronskian. (b) form the linear differential equation for which $y_1(x), y_2(x)$ are solutions and hence find a particular solution with respect to the initial value conditions $y(0) = 2,\ y'(0) = -2$	10	CO2
8	Solve the Non-homogeneous ordinary differential equation $y'' + 3y = 18x^2, y(0) = -3,\ y'(0) = 0.$	10	CO2

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create